

SPECIAL ISSUE ARTICLE

Informed Consent for Research Participation During Space Exploration: Ethical Issues

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ABSTRACT

While informed consent is widely recognized as a cornerstone of ethical research with human participants, applying this principle to space exploration research creates some unique challenges and dilemmas. In this article, we examine ethical issues related to informed consent for participation in space exploration research and offer some suggestions for policy development and further investigation. We argue that: (1) informed consent for space exploration research should be required, but exceptions can be justified for research involving public records, de-identified medical records, secondary use of de-identified, biospecimens or data, emergency medical research, and research critical to mission success or crew safety; (2) organizations that conduct space exploration research should take steps to minimize the potential for undue influence by, for example, ensuring that informed consent discussions will be private and that information about research participation will be kept confidential; (3) prospective spaceflight participants should be provided with information about research activities or other types of data and specimen collection (such as quality assurance) that will take place; and (4) since existing laws and regulations do not apply to most of the research activities conducted by private companies, space companies should give serious thought to developing guidelines for the protection of human research participants.

1 | Introduction

Informed consent is a fundamental principle for ethical research with human participants [1–3]. It is the first principle of the Nuremberg Code and is featured prominently in international guidelines and national laws concerning research with human participants worldwide [4, 5]. Despite abundant guidance concerning legal and ethical requirements for obtaining and documenting consent, controversies about informed consent persist. This is especially true of consent for participation in research where there is high scientific uncertainty and medical or psychosocial risk.

In this article, we examine ethical issues related to informed consent for participation in human research in space and offer

some suggestions for further investigation and policy development in this area. Reflection on this topic is important for several reasons. First, scientific curiosity with, and significant federal investment in, human space exploration has a long history. Tragic failures in the pursuit of safe human spaceflight have resulted in loss of human life, accentuating the importance of understanding spaceflight-associated risks and mitigation strategies [6]. Second, to maintain public trust in human space exploration, both government and industry-supported space programs must adhere to the highest ethical standards, especially standards relating to consent [7]. Although national agencies have provided some guidance on informed consent for flight selection, guidance is far more limited on consent to

participate in a research study conducted during that mission [8–10]. Third, informed consent for research participation in the context of commercial spaceflight, in particular, raises new questions. For example, is there a social responsibility to participate in research? When, if ever, can consent for research participation be waived? And how, if at all, might the benefits of spaceflight create an undue influence on consent decisions?

Before examining these, and other, ethical questions, we will first review factors unique to human research in space as well as provide real-world examples of current studies to facilitate understanding of consent challenges in context.

2 | The Changing Landscape of Human Space Exploration

The people selected to travel to space, as well as the means by which they get there, have evolved dramatically since the launch of the first government space programs. Government astronauts¹ flying the first human space missions were exclusively young men who met stringent medical and psychological qualifications and were current or former military test pilots. The National Aeronautics and Space Administration (NASA) admitted astronauts with more diverse expertise without a military background (e.g., scientists and engineers) for the Space Shuttle missions in the 1970s [11]. In 1985, NASA selected a school teacher, Sharon Christa McAuliffe, as flight crew member for the Space Shuttle Challenger, which tragically was destroyed 73 s after takeoff on January 28, 1986 [12]. Nearly three decades later, SpaceX launched the first all-civilian orbital human space flight in 2021 and a new era of commercial spaceflight was born [13]. Today, passengers as old as 90 and those with pre-existing health conditions have flown with companies including Virgin Galactic, SpaceX, and Blue Origin [14].

Another challenge is that regulation of the commercial spaceflight industry is in flux. Congressionally issued moratoriums on regulation for spaceflight have been in effect for decades. No single federal regulatory agency oversees the safety of crew members and passengers aboard commercial spaceflights. The lack of regulatory oversight means current protections for astronauts, including government and non-government spaceflight participants, are not uniformly applied across the commercial spaceflight industry. Nor are such protections, including standards for informed consent, widely shared to ensure the welfare of all commercial crew members or spaceflight participants regardless of the company who flies them.

We wish to draw attention to an important distinction between consent for *spaceflight*, and consent for *research* in spaceflight. While a moratorium exists for regulation of spaceflight, regulations for human research in commercial spaceflight could conceivably and—as some of us argue elsewhere—ought to be developed. At present, commercial companies that do not accept federal funds are not legally bound by the requirements for human research protections outlined in the Common Rule, even though it is in the company's best interest to follow these standards as a matter of ethical practice. The industry is thus practicing self-governance with respect to consent given there is no central, external authority to enforce consent standards at present.

Historically, flight crew for government-funded research missions (or the government in partnership with a private company) were primarily government employees or military personnel. The purpose of these research missions were ostensibly to benefit the public. Information collected about crew members benefitted future spaceflight participants and contributed to our general understanding of the effects of space travel on human health. Scholars have therefore argued that government astronauts will continue to have a social responsibility to participate in research to further expand the evidence base for safe spaceflight. Although government astronauts often accept the idea that their duty to participate in research stems from an occupational responsibility, it is worth noting that some have refused to participate [15].

Lastly, many early commercial spaceflights fly to low-Earth Orbit (LEO), but this may soon also change. As part of the Artemis missions, NASA plans to send the first humans back to the moon in over 50 years. Both NASA and commercial companies have set long-term goals of reaching Mars, and NASA has funded exploratory research on different propulsion systems that could be used to travel within the solar system or beyond [16–18].

3 | Challenges of Space Exploration

While the regulatory and ethics landscape of safe human spaceflight is evolving quickly, there are many known physical, medical, and psychological challenges of space exploration that have ethical implications.

Radiation: On Earth, the atmosphere and magnetosphere protect life from the harmful effects of ionizing radiation emitted from the sun and the cosmos. Astronauts that travel outside Earth's magnetosphere (which extends beyond the moon) are exposed to increased radiation. NASA is developing technologies to block or deflect radiation, but radiation will always be a major hazard for space travel [19, 20].

Zero Gravity: Individuals who fly to space are exposed to zero-gravity environments. Gravity on the moon is 16% that of the gravity on Earth. On Mars, gravity is 38% that of Earth, and it can reach close to 0% in interplanetary space. Prolonged exposure to low gravity accelerates bone loss, muscle atrophy, and other medical problems. While NASA is considering ways of generating artificial gravity, for example, by spinning a spaceship or space station to produce centrifugal force, low gravity will always be a health-related concern for prolonged human presence in space [19, 21].

Isolation and Confinement: People who fly to space are isolated from the rest of society during their time in space and confined to a small space with each other. The International Space Station (ISS) is designed to comfortably house six people. For the foreseeable future, crew sizes on missions to the moon or Mars are not likely to be much larger. Whether the size of the crew is 6 or 60, people who venture into space will have to manage the psychological stresses of isolation and confinement [19, 20].

Hostile Environments: The environment outside the spaceship or space station does not support human life, due to lack of a breathable atmosphere, radiation, and excessive or insufficient pressure or temperature. The hostile external environment increases confinement because even when individuals land on a

planetary body they will not be able to permanently escape the safety of their artificial environment.

Limited Resources: On a spaceship or space station, material and human resources such as food, water, energy, medicine, and expertise/skill are extremely limited. These limitations necessitate rationing of material resources and versatility of human resources; for example, the same person may serve as the ship's physician, surgeon, dentist, psychologist, epidemiologist, and clinical researcher.

The challenges of space exploration also have ethical implications. Space exploration is an inherently dangerous, isolated, and confined activity in which the survival of the group is paramount. As a result, the interests of the group may legitimately override those of individuals in some circumstances.

4 | Current Human Studies Involving Space Exploration

Numerous studies published in the scientific literature have investigated the effects of space flight on human health [22, 23]. Most of these have an observational as opposed to experimental study design.² For example, Lee and collaborators collected and analyzed magnetic resonance imaging (MRI) scans from 26 astronauts taken prior to and after spaceflight, including 7 who were on short-duration Space Shuttle flights and 8 who spent a longer time (200+ days) on the International Space Station (ISS). The authors found that spaceflight was associated with changes in the distribution of extracellular free water in astronauts' brains and that some of these changes were associated with duration of time in space or the number of spaceflights [24]. Similarly, Inglesby and collaborators examined the pre- and post-spaceflight MRI scans of 35 astronauts, including 17 who were on Space Shuttle missions only and 18 who were on ISS, and found exposure to spaceflight conditions on ISS was associated with an increased likelihood for the formation of mastoid effusions [25]. Mehta and collaborators collected blood, urine, and saliva specimens from 17 astronauts prior, during, and after spaceflight; tested the specimens for antiviral antibodies and viral load DNA for Epstein-Barr virus, varicella-zoster virus, and cytomegalovirus; and found that all three of these viruses reactivated during spaceflight [26]. Because this study involved specimen collection while in space, the astronauts had to assist with it, which raises issues concerning the participant's role as a co-investigator, which we will not explore here [27].

Although there have not been many clinical trials that test products for use in space flight, this is likely to change as the population of astronauts increases. For example, Defender Pharmaceutical plans to recruit volunteers from commercial and government space missions for a clinical trial of a medication to treat nausea due to motion sickness [28]. Researchers at Charité University, Berlin, Germany, are planning to test a non-significant risk medical device on 8 astronauts on ISS. The device electrically stimulates muscles to optimize the physiological effects of exercise [29]. Researchers at several institutions are collaborating with the Translational Research Institute for Space Health to fund and conduct scientific studies and technologies to protect the health and safety of astronauts [30]. Clinical trials in space pose significant challenges due to the small sample sizes and unknown risks of how interventions may differ when

performed in space as compared to on Earth. Another challenge is controlling the experimental conditions of the study, such as radiation exposure or length of time space [31, 32].

5 | Consent for Research Participation Versus Consent for Participating in a Space Mission

In 2014, the National Academy of Sciences, Engineering, and Medicine issued a report that provides valuable guidance for informed consent to participate in a space mission. According to the report, to be ethical, a proposed space mission must "have a rigorous consent process to ensure that astronauts are fully informed about risks and unknowns, meet standards of informed decision making, and are making a voluntary decision." [7, p. 9] While this report addresses informed consent for participating in a space mission, it has little to say about informed consent for participating in research done during a space mission. This is an important distinction that is similar to the distinction, well documented in the research ethics literature, between consenting to a medical procedure and consenting to participate in a study conducted as an adjunct to that procedure [33]. As noted above, space exploration can be a high-risk activity fraught with uncertainty. While research procedures conducted during or after the mission, such as specimen collection or imaging studies, may themselves be low-risk, these risks may increase in space environments. Uncertainties concerning the risks of being exposed to ionizing radiation, for example, could lead to uncertainties concerning the risks of an imaging study conducted on Earth after flight since the risks of ionizing radiation are cumulative [34].

Another relevant consideration is the distinction between data or specimens collected for quality assurance/quality improvement (QA/QI) purposes related to space missions versus data/specimens collected for the purposes of research. QA/QI is conducted to help an organization (such as a hospital) improve its operations [35]. Research, in contrast, develops generalizable knowledge that could benefit people beyond the scope of the organization conducting the research [36]. Consider a space agency or private company that collects data on its CSPs to monitor their health and safety but not as part of a research study. Consent for QA/QI is not generally required since it is assumed that consent for participating in an organization's operations implies consent for its QA/QI activities. For example, a hospital does not need to ask patients whether it can collect data on the effectiveness of its infection control practices since these are understood to part of its operations. Likewise, a government agency or private company does not need to obtain consent for QA/QI activities related to protecting the health and safety of its astronauts. The difference between QA/QI and research can be murky, however, since methods used for collecting data or specimens for QA/QI purposes might yield generalizable knowledge or data or specimens collected for QA/QI might subsequently be used in a research study. For this reason, it is advisable for an organization to consult with its IRB concerning QA/QI activities [36].

6 | Should Consent for Space Exploration Research Always Be Required?

The first issue we will consider is whether informed consent for space exploration research should always be required. Many

studies of human participation in space exploration do not involve consent because they rely on data from public records or astronauts' de-identified medical records, both of which are considered to be exempt research under the Common Rule.³ Of course, the fact that these studies have been legally exempted from IRB review does not settle the ethical question of whether consent ought to be required. One could argue that the studies should not have been exempt in the first place, or one could challenge the legitimacy of some of the exemption categories. For example, many people regard the use of Henrietta Lacks' leftover tumor tissue without her consent as unethical, even though the use of leftover samples could qualify as exempt if it took place today [37, 38]. We will not question the exemption determinations for the studies mentioned above nor critique the exemption categories. Rather, we will consider how ethical arguments for not requiring consent may apply to human research in space.

Although the principle of respect for autonomy provides a compelling justification for an absolute right to decide whether to participate in research, some have argued that this right can be restricted for important research that imposes only minimal risks on participants [39]. There are several types of research for which consent is not usually required, including research involving de-identified medical records, secondary use of de-identified data or biospecimens, and hospital quality improvement research.⁴ The standard ethical argument for not requiring consent in some situations is supported by three premises [39]:

1. The research is socially valuable;
2. The risks are minimal;
3. The research would be impracticable (i.e., very difficult to do) if consent from each individual in the study had to be obtained.

Should these premises likewise apply to human research in space? Premise (1) clearly applies to most studies involving *government* astronauts. The research is socially valuable because it is likely to benefit the health of other astronauts and facilitate the success of future missions. The fact that research helps expand the limited evidence base for spaceflight in the future justifies a common practice that all those who will fly to space should be offered the opportunity to participate in research [40].

The minimal risk premise is more ambiguous. The risks of research procedures, such as imaging studies, biospecimen collections, and physical examinations, are likely to be minimal in many, but not all, cases. For example, drug or device trials done to protect the health of astronauts would likely be more than minimal risk. Also, some observational studies may involve more than minimal risk procedures, such as biopsies, cytophoresis, or some types of medical imaging. Since the risks of radiation exposure are cumulative, a procedure normally classified as minimal risk, such as a chest x-ray, may pose a greater risk for a person who has already been exposed to considerable radiation in space [41]. Additionally, the risks associated with loss of privacy/confidentiality, such as discrimination or stigma, may not be minimal [42]. The loss of privacy/confidentiality is a major concern because it may be difficult to prevent outside parties from re-identifying

participants [42]. Because the pool of astronauts involved in a particular space exploration study is likely to be very small (e.g., $n < 10$), it may be relatively easy to re-identify individuals in de-identified datasets, especially if genomic information is included. Moreover, closely-related family members may also be impacted by the loss of privacy/confidentiality, since it may be possible to infer that astronauts and family members are likely to share common genetic conditions [42].

Premise (3) usually applies to situations in which consent would be impracticable based on the excessively large number of participants or if the consent process would affect the scientific integrity of the study [43]. Precedent for consent waivers based on the impracticability premise does not seem to apply to space exploration research, given the small size of the participant pool. Obtaining consent from a small number of individuals in advance of a mission would be practical and is done frequently in other types of small-N studies in rare disease research, for example. The practicality of consent may diminish, however, as the total N increases. For now, the claim that the research would be impracticable if consent were required does not stand up to close scrutiny. One could also take a very different slant on the impracticability premise by arguing that consent should not be required because the participant pool is so small that if only a few people decide not to participate, the study would be invalid due to the lack of statistical power. However, this argument is not very convincing because a high percentage of astronauts are likely to participate since they understand the importance of the research.

In 2001, authors of an Institute of Medicine report on space medicine developed a different argument for waiving consent for space exploration research. The authors proposed that an "occupational health model" should apply to space travel. That is, having the occupation of being an astronaut should carry with it strong obligations to promote the health of the occupation, including responsibilities to participate in research.⁵ [9] One problem with this approach, however, is that society does not recognize this responsibility for many other healthcare occupations (e.g., medicine, nursing, hospital administration) and, therefore, human research regulations do not allow an IRB to waive consent requirements for healthcare professionals who are asked to participate in research. To avert this objection, the authors of the IOM report argue that space travel entails added responsibilities because of its uniqueness:

The opportunity to observe the effects of space travel on humans is remarkably rare and thus is a scarce resource that should be used or allocated in the most efficient manner possible... Such an opportunity brings with it the responsibility to give up some privacy [and autonomy] to ensure the safety of fellow astronauts and those astronauts who will follow.

[9, pp. 178–179]

Although this passage refers to sacrificing some degree of "privacy" one could interpret this argument to apply to other individual interests more broadly. Building on the IOM report, we propose that:

1. The opportunity to observe the effects of space travel on human health is a valuable and scarce resource;

2. Valuable, scarce resources should be used wisely and responsibly;
3. To help ensure that this resource is used wisely and responsibly, humans that travel in space (i.e., astronauts) have social responsibilities, which could include agreeing to some loss of privacy or other trade offs of participating in research that expands the evidence base.

We agree with the authors of the report that the opportunity to observe the effects of space travel is a valuable, scarce resource that should be used wisely. We also suggest that individuals who travel into space, including not just professional astronauts and scientists but also tourists, have social responsibilities to participate in research, given their unique opportunity to do something that so few people in the general public could also realistically do now [40]. However, acknowledgment of this social responsibility does not imply that consent can be waived for all space research, because the argument that supports this inference would justify waivers of consent in other situations in which people have social responsibilities (e.g., research on individuals with rare diseases, those in isolated environments, disaster victims, or military personnel exposed to unique risks), and applying a waiver to these cases is ethically problematic as it may lead to research exploitation and coercion.

Although we strongly support requiring informed consent for space research, we recognize that there should be some exceptions which are consistent with accepted practices for human participation in terrestrial research. First, consent of the astronaut need not be required for certain types of low-risk research in which consent is not normally required, such as research on public records, de-identified medical records, or secondary research involving their de-identified data [42]. As noted above, the scientific literature already contains many examples of this first exception.

Second, an astronaut's consent may not be required if they are facing a medical emergency and are unable to provide consent. FDA regulations cover two types of emergency situations: emergency use of a test article (or emergency medical treatment) and emergency research. The FDA regulations permit the use of unapproved test articles (such as drugs or devices) to treat patients if the following conditions are met [44, 45]:

1. A person is facing a life-threatening medical emergency;
2. There are no effective treatments for their medical problem;
3. The product being tested may be effective at treating the problem;
4. The person is unable to consent (e.g., because they are unconscious); and
5. Their legally authorized representative is not readily available.

Under these conditions, a physician can use an experimental product to treat patient without obtaining consent from the person or their legally authorized representative or receiving permission from an IRB. The rationale for not requiring consent under these emergency conditions is that most people would want to receive potentially life-saving medical treatment. Prior IRB approval is not required because there is not enough time

to obtain approval in the face of a life-threatening emergency. However, the investigator must notify the IRB within 5 days of the emergency use [45].

FDA regulations allow the IRB to waive informed consent requirements for emergency research if all four of the conditions listed above are met as well as several others, such as prior consultation with the community about the proposed study and evidence for safety and effectiveness for the product being tested from prior animal and laboratory studies. IRB approval is required to do emergency FDA-regulated research. Although the Common Rule does not include provisions for IRB waiver of consent for emergency medical research, the Department of Health and Human Services has waived consent requirements for emergency research that meets the conditions set forth in the FDA regulations as well as other Common Rule requirements [44].

At present, it is highly unlikely that a situation would arise during space travel in which emergency use of an investigational product or emergency research would be appropriate. However, it is conceivable that the need for emergency medical research could arise in the future. For example, suppose that an individual on a moonbase has a life-threatening medical emergency, such as decompression sickness from a malfunctioning spacesuit. Standard treatments are not effective at treating their condition and the only way to save their life is to give them a treatment that has been tested with animals on Earth but not in humans. The individual is also incapable of making an informed choice about receiving the experimental intervention and their legally authorized representatives are on Earth and cannot be readily contacted. We think it would be appropriate to waive the strict requirement for consent in a case like this, either as an emergency use of a product or as emergency research. Again, we do not regard this conclusion as unusual or controversial, since it is consistent with how human research participants are treated on Earth.

We also contend that participants should also retain their right to withdraw consent to participate in research before, during, and even after the research mission. Cost, while potentially consequential for mission donors/sponsors in the commercial context, should not on its own factor into a participant's right to withdraw. Rather, withdrawal becomes more ethically contentious in cases where it compromises the safety or creates new ancillary risks to the participant or the rest of the crew that were not anticipated or agreed to at the time of consent.

A more controversial case would involve forcing someone to receive an experimental treatment or participate in a research study for the sake of their crew members and perhaps the mission itself. Suppose that during a flight to Mars a deadly respiratory pathogen has infected two crew members. One died quickly after displaying symptoms and the second is only mildly ill. To prevent the disease from spreading and optimize treatment for sick crew members, it is necessary to better understand the pathogen, and to accomplish this goal data and biospecimens must be collected from the living crew member. However, the second crew member refuses to comply with requests to do this. By refusing to give consent for data and specimen collection, the crew members jeopardize the lives of other crew members and the mission itself.

This case blurs the line between research and other activities involving data and specimen collection—in this situation,

public health (and possibly QA/QI as well). The data and specimens are collected, first and foremost, to save the lives of other crew members and secondarily to contribute to scientific knowledge. Therefore, one might argue that this case is better understood as a form of public health practice and not as a type of research [46]. However, we think the matter is best viewed as involving public health and scientific investigation because the research objectives, though secondary, may still impact data and specimen collection activities. For example, the physician might collect more specimens than are needed for public health purposes so that some will be available for research.

Although we are claiming that overriding consent is justified to save the lives of other crew members, we are not claiming that it applies to achieving longer-term research goals. For example, data and samples could be collected and used without consent to do the research necessary to save the lives of people but could not be used for future studies unrelated to the emergency situation without consent.

Regardless of whether one views this as case as public health practice, research, or both, consent issues will still be paramount. It may be acceptable to violate a crew member's right to refuse to consent for data and specimen collection to prevent serious harm to other crew and the mission itself. This rationale would be consistent with principles of public health ethics that justify coercive measures, such as mandatory treatment or restrictions on movement, in situations involving serious threats to public health [47].

A growing mix of both commercial and government astronauts now comprise crew of research missions. Within the group of commercial astronauts, some are employees of the company and others may be customers. Separate standards for informed consent based on prior spaceflight experience become problematic with mixed crews. Our position, which maximizes fairness and respect for persons, is to favor a common, reasonable standard for informed consent for all current and future spaceflight participants, including government astronauts, commercial spaceflight company employees and other civilians who may fly to space.⁶ Attributes of fully informed consent to participate in research on Earth (i.e., that capacity to consent is establish, the decision to participate in voluntary and free of coercion) are transportable to informed consent for research participation in the spaceflight context. While extending federal human research protections to the commercial spaceflight industry requires a Congressional vote, alternative paths could involve the voluntary adoption of ethical standards by industry leaders, private rights of action through extant liability and tort mechanisms, and/or defaulting to NASA standards and procedures until industry-specific guidance is developed and enforced.

7 | Should Astronauts Be Asked to Consent to Space Exploration Research That May Take Place in the Future?

One strategy for alleviating some of the moral conflict related to the cases described above would be to ask individuals on space missions to consent to future (and potentially unknown) collection of data and biospecimens for research and QA/QI, and to consent to medically indicated uses of experimental products.

Individuals could also be informed that since spaceflight has heightened risks and uncertainty, it is impossible to anticipate all the types of situations that may involve their participation in these activities, but that situations could arise in which their participation would be vital to protecting their own life, the lives of others, and the success of the mission. This strategy would operationalize the IOM report's public health model by making the nature of the mission explicit, and would be similar to obtaining consent to future, as yet unspecified, uses of data or specimens from human research participants [42, 48].

While we think this strategy is useful for informing astronauts about some of the challenges and opportunities that may arise during a space mission and promoting transparency and trust, it does not constitute valid consent because it fails to provide enough information about participation in future research, experimentation, or data and specimen collection. Individuals would be consenting to much more than undetermined, low-risk uses of specimens and data that have already been collected but also new, and possibly risky, collection activities, experiments, and interventions. With the exception of a few types of situations mentioned above, consent for research should be sought from the participant or their legal representative prior to enrollment in a study, even when they have already been given advance notice that research may take place.

8 | How Can Coercion and Undue Influence Related to Space Exploration Research Be Minimized?

In its discussion of informed consent, the IOM report notes that:

[T]he research review process must recognize the unique pressure on astronauts to overcome any obstacle to participating in space travel. The usual methods of informed consent...fail to recognize the intense competition among astronauts to be assigned to a particular mission and the implicit coercive effects of such competition. Although it is NASA policy that refusal to participate in a research protocol should not influence the crew selection process... there is evidence that the competition for crew positions renders this policy almost meaningless in practice. Astronauts are reluctant to reveal medical incidents or complaints and to report symptoms during space missions for fear that any physiological variation, however small, may jeopardize their chances of future space travel.

[9, pp. 182–183]

The possibility of coercion or undue influence related to space exploration research is a serious issue that must be addressed properly.⁷ Clearly, astronauts competing for positions in a government space program face significant pressures to comply with real or merely perceived requirements. Astronauts working for private companies probably face similar pressures. Although private citizens who fly to space likely will not experience these same pressures, their decision to participate may be unduly influenced by the unique prospect of traveling to space.

Steps can be taken to minimize this influence. Members of the professional government astronaut corps are informed that research participation has no impact on mission selection. The commercial spaceflight companies could adopt a similar policy. Consent from research participants could also be obtained by someone who is not their direct (or indirect) supervisor. Consent discussions should take place in private, which reduces the risk that other crew members will be able to discover who is (or is not) participating in a study. Research records should also be kept confidential. Admittedly, privacy may be difficult to protect in the confined area of a space base, spaceship, or space station, but it should be protected, the extent that it can be [49].⁸

9 | Concluding Discussion

Informed consent is a foundational issue in research with human participants, whether it takes place on Earth or in space. We have examined some of the key informed consent issues for research participation in space that may arise now or at some future time and conclude the following:

1. Due the challenges of extraterrestrial living, such as extreme isolation and interdependence, space exploration research raises some unique consent issues that do not have clear analogs on Earth.
2. In thinking about the ethics of informed consent for research participation, it is important to distinguish between:
 - a. consenting to go on a space mission vs. consenting to participate in space mission research and
 - b. research, QA/QI, medical treatment, and public health practice.
3. Informed consent for participation in space exploration research should be required except for
 - c. research on public records;
 - d. research on de-identified medical records;
 - e. secondary use of de-identified data or biospecimens collected for other research purposes;
 - f. Emergency medical research; or
 - g. Research vital to protecting the lives of other crew members or the success of the mission.
4. Since people may feel social pressure to participate in research, organizations that conduct space exploration research should take steps to minimize the potential for undue influence by, for example, ensuring that informed consent discussions will be private and that information about research participation will be kept confidential.
5. Providing prospective spaceflight participants with information about the types of situations that may arise during their mission that may involve research, experimentation, QA/QI studies, or data and specimen collection can help to reduce moral conflict and psychological stress when questions about research participation arise, but do not constitute, valid informed consent.
6. Further investigation of the ethics of informed consent for space exploration research is urgently needed, including philosophical, legal, and policy analysis and

empirical studies involving surveys, interviews, and focus groups.

In closing, we would like to note that we have included private space companies in our discussion even though existing human research protection laws and regulations do not apply to their activities, unless they are collaborating with or working for a government agency or doing research to support an application to the FDA. Although international ethical guidelines, such as the Helsinki Declaration and CIOMS guidelines, can provide some useful guidance for private companies, space companies currently operate largely beyond the scope of the human research protection laws. We therefore urge space companies to give serious thought to developing guidelines for the protection of human research subjects to enhance their public image, minimize their liability under tort law, and because it is the right thing to do.

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Conflicts of Interest

The authors declare no conflicts of interest.

Endnotes

¹We use the term "astronaut" in a broad sense to refer to any human being travelling in space, including professional astronauts (whether employed by the government or private industry), scientists conducting experiments in space, and tourists on commercial space flights.

²Observational studies collect data on people without directly manipulating the conditions whereas experiments manipulate the conditions to determine whether specific variables (such as taking a drug) predict specific outcomes (weight loss). For example, collecting on data on the long-term risks of taking a weight loss medication would be an observational study but conducting a controlled clinical trial to learn about the benefits and risks of the medication would be an experiment.

³The Common Rule (45 CFR 46) is a regulation adopted by 17 federal agencies for protection of human subjects. NASA follows the Common Rule. Its version of the rule is at 14 CFR 1230 [8].

⁴Secondary use refers to use of data or specimens that have already been collected for another study. For example, if an astronaut consent to participate in a study, a secondary use would occur if their data or specimens were used for a different study [42].

⁵Although NASA commissioned the IOM report, it did not adopt the IOM's recommendations concerning informed consent.

⁶Although the standards of consent should be the same for all astronauts, the process of consent involving employees may need additional oversight to minimize the potential for coercion or undue influence [42].

⁷Coercion involves a threat of physical, psychological, financial, or legal harm, where undue influence does not. Undue influence occurs takes advantages of someone's else vulnerability or ignorance to induce them to make a decision; for example, a psychotherapist having a sexual relationship with a patient [42].

⁸The medical confidentiality problems on a spaceship are similar to those that arise in Antarctic stations [48, 49].

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